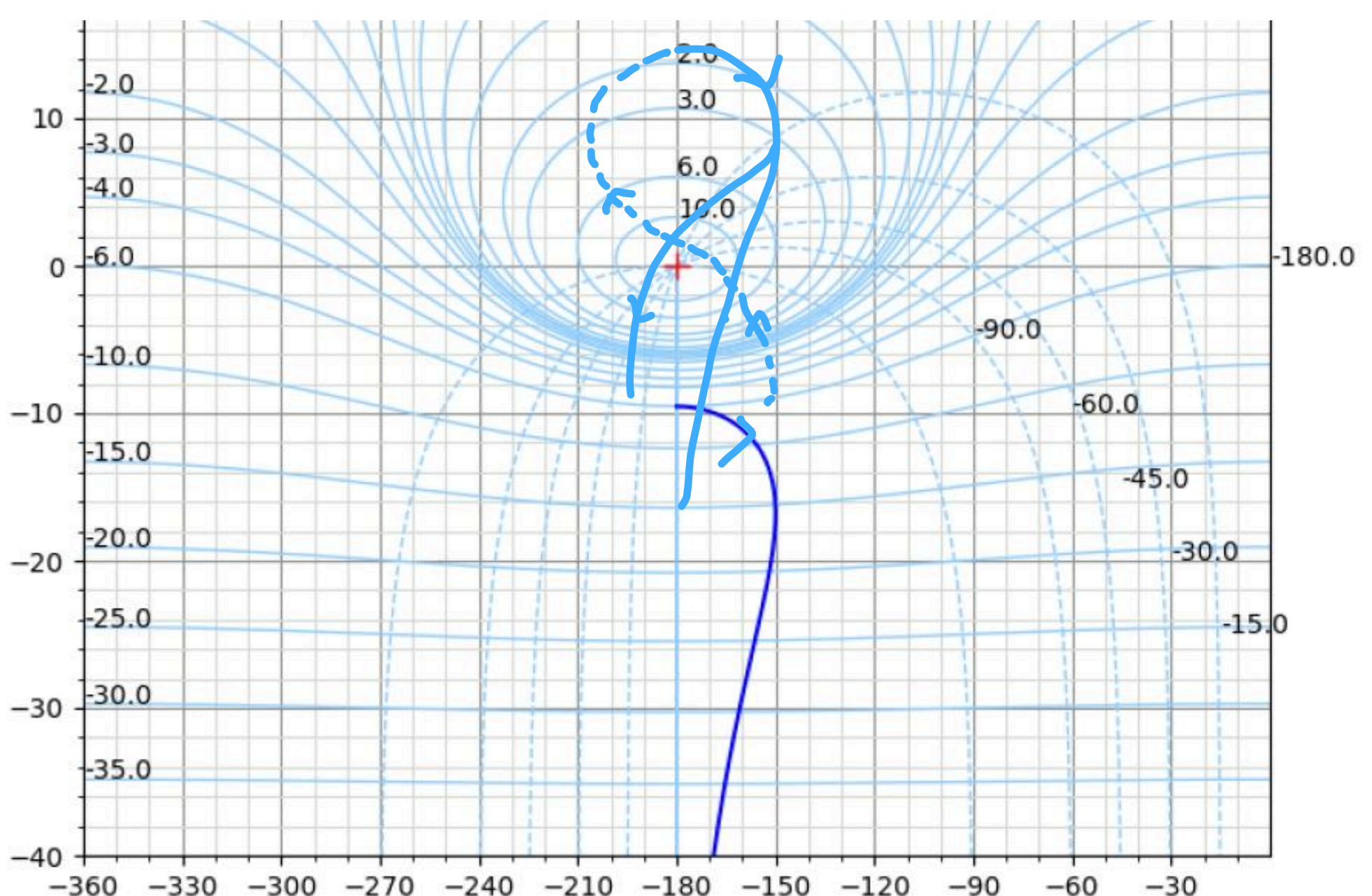
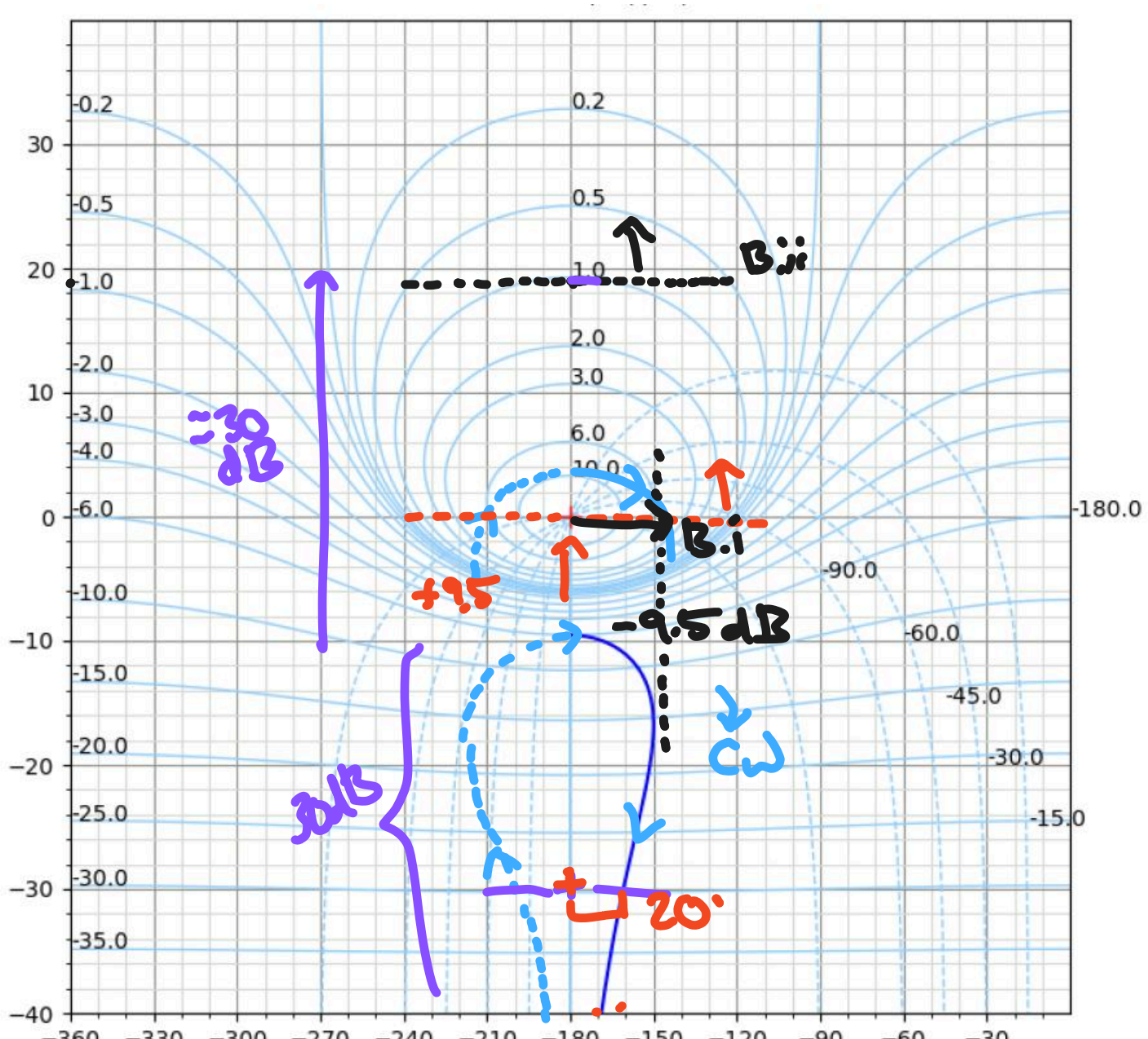


Test 2 Breakdown

Question 3 [13 marks]

The Nichols chart of the unstable system $G(s) = \frac{1}{(s-1)(s+3)}$ is shown below.

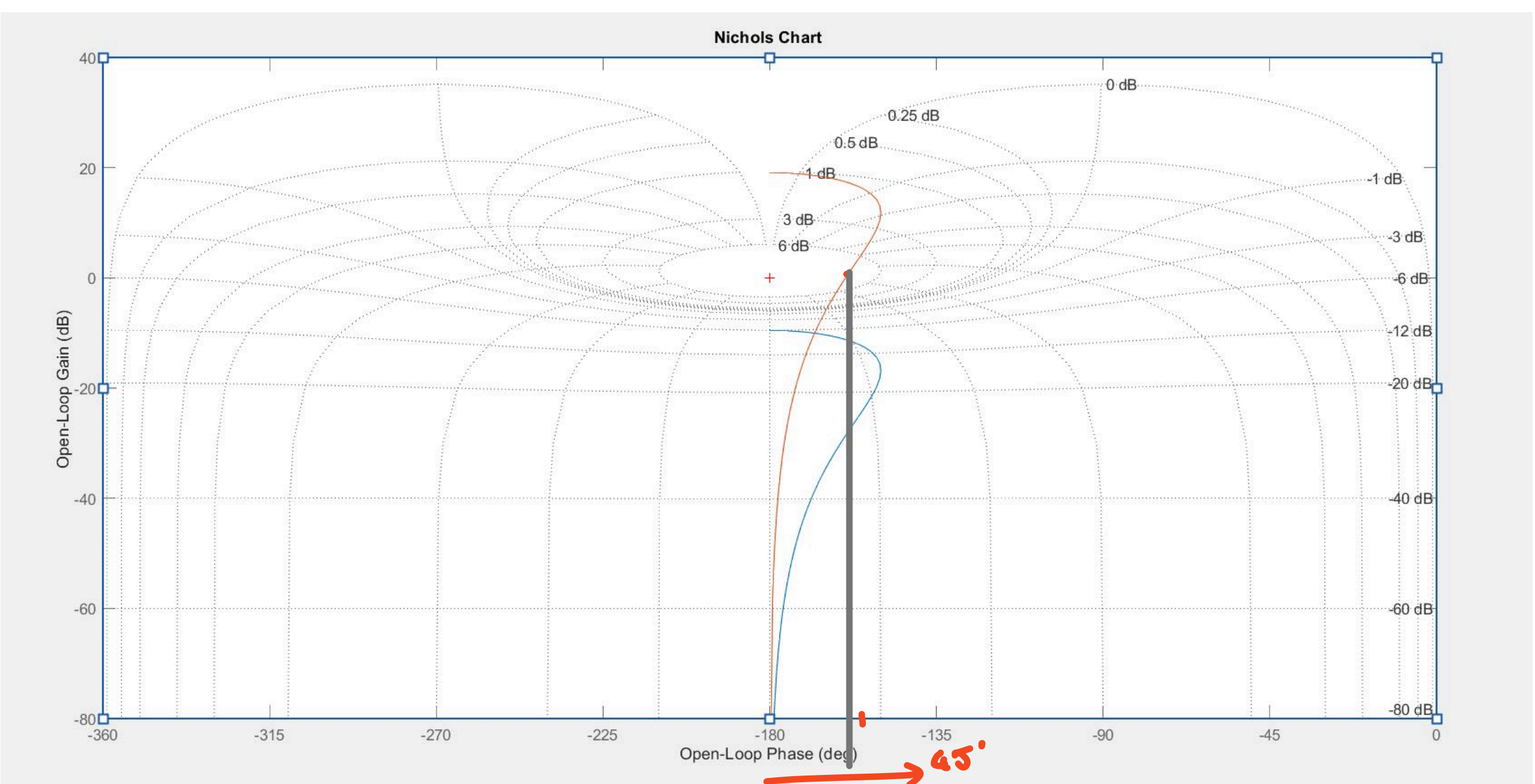
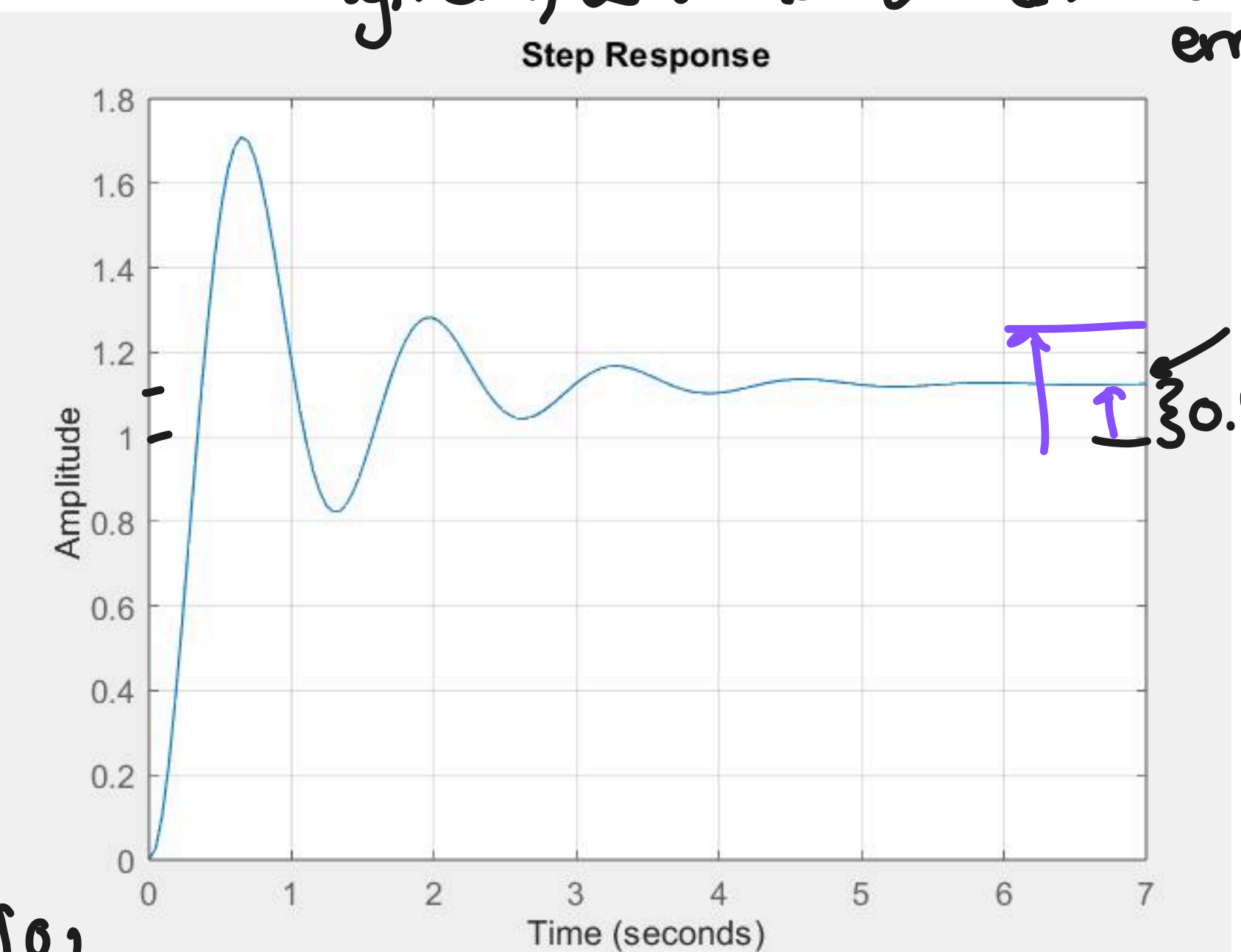
- A) With reference to the Nichols chart, explain how you know it is possible to stabilize the system by placing it in negative feedback with a proportional controller, and determine the range of controller gains for which the closed-loop system will be stable. Indicate this on the Nichols chart by drawing a specification boundary. [4 marks]
- B) Draw specification boundaries on the plot representing the following:
- i) A phase margin of 30 degrees [1 mark]
 - ii) Tracking accuracy of 90% with respect to constant setpoints. [2 marks]
- C) Is it possible to meet both the specifications in part B at once? You may draw on the plot to support your explanation. [3 marks]
- D) If you prioritize the accuracy specification in part B, ignoring all other specifications, give a best-case approximation for the percentage overshoot of the closed loop system. [3 marks]



$$N = P_{cl} - P_{ol}$$
$$1 = 2 - 1$$

$K=27$
D. $\rightarrow 1e + 28dB$

CL system, with $K=27$ ($\therefore$ satisfying error < 0.1)



$$\lim_{\omega \rightarrow 0} G(j\omega) = \frac{1}{j\omega - 1 \times j\omega + 3} = -\frac{1}{3}$$
$$|G(j\omega)| = 20 \log_{10} \left(\frac{1}{3} \right) = -9.54 \text{ dB}$$
$$G_{cl} = \frac{1}{(s-1)(s+3)}$$

$$N = P_{cl} - P_{ol}$$
$$-1 = 0 - 1$$

(1 CW)
(Nichols)

$\hookrightarrow$ with $K=27$ (28 dB)
get PM $\phi_m = 20^\circ$
(A) $3 \approx \frac{1.1}{100} \cdot 20 \approx 0.22$

$$\therefore \%OS = \frac{e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}}}{\sqrt{1-\zeta^2}} \cdot 100$$
$$= 49\% \text{ OS}$$

FV of y: $\lim_{s \rightarrow 0} s \cdot \frac{\Phi}{1+\Phi} = \frac{\Phi_0}{1+\Phi_0} \rightarrow \Phi_0 = \Phi(0)$
FV of e: $\lim_{s \rightarrow 0} s \cdot \frac{1}{1+\Phi}$

$$\lim_{s \rightarrow 0} s \cdot \frac{K G(s)}{1+K G(s)} \leq 0.1 \rightarrow \frac{1}{1+K \left(\frac{1}{3} \right)} \geq 0.1$$
$$1 - \frac{K}{3} \geq -0.1$$
$$-\frac{K}{3} \geq -1.1$$
$$K \leq 3.3$$

Question 4

A) Does the addition of the integrator affect the gain margin of the system? [3 marks]

B) The aim of placing the system in negative feedback is for it to satisfy the following specifications:

- i) Tracking position setpoints with accuracy of more than 95%.
- ii) Percentage overshoot of less than 20%.

Decide whether the system with the integral controller meets each of these specifications, supporting your answers with features from the Nichols chart. [2 x 3 marks]

C) Your friend thinks it's unnecessary to go through the effort of implementing an integral controller, when it's possible to achieve the required tracking accuracy with a proportional controller.

- i) Determine the minimum gain required of this proportional controller, and re-draw the response

of the original system $G_1(j\omega)$ at this new gain. [2 marks]

(An additional chart is provided at the end of the answer space if you need one.)

ii) Based on the response of the system with the proportional controller, would you agree with your friend that proportional control is the way to go? You may draw on/ annotate the Nichols chart further to support your answer. [3 marks]

No: ϕ_m becomes 25°
 $\therefore$ too much OS.

$$\frac{1}{1+\Phi_0} \leq 0.05 \therefore \Phi_0 \geq 19$$
$$\approx 26 \text{ dB}$$

